

NEETA KHANUJA

PhD Candidate, Human-Computer Interaction Institute, Carnegie Mellon University
and Instituto Superior Técnico, University of Lisbon

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RESEARCH INTERESTS

Human-AI Interaction • Self-Perceptions of Aging • Culturally Situated HCI • Interactive Art and Systems

EDUCATION

Dual-degree Ph.D., Carnegie Mellon University and Instituto Superior Técnico Sep 2021 – Aug 2027

Ph.D., Human-Computer Interaction, Carnegie Mellon University, Pittsburgh, USA

Ph.D., Information Systems and Computer Engineering, Instituto Superior Técnico, University of Lisbon, Portugal

Advisors: Jodi Forlizzi (Carnegie Mellon University), Valentina Nisi (Instituto Superior Técnico)

Dissertation: Operationalizing Self-Perceptions of Aging in HCI Research and Design

M.S., Human-Computer Interaction, Carnegie Mellon University 2022 – 2025

Dual-degree Master's, International Media Architecture Master Studies 2016 – 2018

M.Sc., Media Architecture, Bauhaus-Universität Weimar, Germany

M.S., Architecture, University at Buffalo, State University of New York, USA

Post Graduate Diploma, New Media Design, National Institute of Design, India 2008 – 2011

B.E., Computer Science, Gujarat University, India 2003 – 2007

PUBLICATIONS

Peer-Reviewed Papers and Pictorials

- [1] **Khanuja, N. M.**, Nisi, V., Meena, Y. K., & Forlizzi, J. (2026). Designing culturally grounded reflection cards that explore self-perception of aging and automated recommendations with older adults in India. In *Proceedings of the ACM Designing Interactive Systems Conference (DIS '26)*, 3202–3215. Pictorial. [DOI](#)
- [2] Gonzalez, J. T., **Khanuja, N. M.**, Li, M. M., Guo, M., Olaitan, L., Lau, E., Pugh, J., Ion, A., & Hudson, S. E. (2026). Towards fluent interaction with cyber-physical architecture. In *Proceedings of the ACM CHI Conference on Human Factors in Computing Systems (CHI '26)*. [DOI](#) (**ACM CHI Best Paper Award**)
- [3] Simão, H., Gonçalves, D., **Khanuja, N. M.**, Nisi, V., Bernardino, A., Guerreiro, T., & Forlizzi, J. (2026). Values across contexts: Understanding how older adults enact what matters through technology. In *Proceedings of the ACM CHI Conference on Human Factors in Computing Systems (CHI '26)*. [DOI](#)
- [4] Chang, M. L., Reig, S., Lee, A., Huang, A., Simão, H., Han, N., **Khanuja, N. M.**, Mohammad Ali, A. U., Martinez, R., Zimmerman, J., Forlizzi, J., & Steinfeld, A. (2025). Unremarkable to remarkable AI agent: Exploring boundaries of agent intervention for adults with and without cognitive impairment. *Proceedings of the ACM on Human-Computer Interaction*, 9(2), 1–26. [DOI](#)
- [5] **Khanuja, N. M.**, Nisi, V., & Forlizzi, J. (2025). Designing aging reflection probes to elicit self-perception of aging (SPA) beliefs of older adults in India. In *Proceedings of the ACM Designing Interactive Systems Conference (DIS '25)*, 2602–2606. Pictorial. [DOI](#)
- [6] Chang, M. L., Lee, A., Han, N., Huang, A., Simão, H., Reig, S., Mohammad Ali, A. U., Martinez, R., **Khanuja, N. M.**, Zimmerman, J., Forlizzi, J., & Steinfeld, A. (2024). Dynamic agent affiliation: Who should the AI agent work for in the older adult's care network? In *Proceedings of the ACM Designing Interactive Systems Conference (DIS '24)*, 1774–1788. [DOI](#)
- [7] **Khanuja, N.**, Freire, M., Lameiras, J., Nicolau, H., Forlizzi, J., & Nisi, V. (2023). Using cultural probes to understand students' mental wellbeing. In *Proceedings of IASDR 2023*, Milan, Italy. Pictorial. [DOI](#)

Under Review

- [8] **Khanuja, N. M.**, Simão, H., Nisi, V., & Forlizzi, J. Understanding aging in urban India: Cultural anchors and directions for design. Under review at ACM CHI 2027.
- [9] **Khanuja, N. M.**, Nisi, V., Gonzalez, J. T., & Forlizzi, J. Technology design opportunities for positive self-perceptions of aging (SPA) in urban India. Under review at ACM CHI 2027.

Extended Abstracts and Doctoral Consortium

- [10] **Khanuja, N. M.**, Meena, Y. K., Nisi, V., & Forlizzi, J. (2026). Operationalizing self-perceptions of aging using reflection probe and SPA design cards with HCI practitioners. In *Extended Abstracts of the ACM CHI Conference on Human Factors in Computing Systems (CHI EA '26)*. [DOI](#)
- [11] **Khanuja, N. M.** (2025). Designing technological interventions to enhance self-perceptions of aging (SPA) and well-being of older adults in India. In *Companion Publication of the ACM Designing Interactive Systems Conference (DIS '25 Companion), Doctoral Consortium*, 111–115. [DOI](#)

Earlier Publications and Presentations

- [12] **Khanuja, N.** (2020). A study of inclusion in smart cities. *AMPS Rapid Cities – Responsive Architectures Conference*, Dubai.
- [13] **Khanuja, N.** (2020). *Designing a way-finding and environmental learning tool for people with mild to moderate dementia* [Master's thesis]. University at Buffalo, State University of New York.
- [14] **Khanuja, N.** (2019). Exploring the potential of universal design in adaptive architecture. *Student Interaction Design Research Conference (SIDeR '19)*, Muthesius University of Fine Arts and Design, Kiel, Germany.
- [15] **Khanuja, N.** (2019). Lost in place: A case study of an interactive installation for dementia awareness. In K. Niedderer, C. Wölfel, G. D. S. Ludden, & R. Cain (Eds.), *Designing with and for People with Dementia: Wellbeing, Empowerment and Happiness. Proceedings of the International MinD Conference 2019*, Dresden, Germany. MinD & TUD Press. ISBN 978-3-95908-183-2.
- [16] **Khanuja, N.** (2018). Design as research and research as design: A case study of a smartphone-based braille-conductive interface. *INSIGHT 2018*, National Institute of Design, Ahmedabad, India. [Link](#)
- [17] **Khanuja, N.**, & Steinfeld, E. (2018). Designing a way-finding tool for people with cognitive difficulties. *RESNA Annual Conference 2018*, Washington, DC. [Link](#)

RESEARCH EXPERIENCE

Doctoral Researcher, Carnegie Mellon University and Instituto Superior Técnico 2021 – Present

- Study how self-perceptions of aging (SPA) can be understood and operationalized in HCI research and design.
- Designed a seven-day Aging Reflection Probe built on established SPA measures and used it with 24 older adults across four living contexts in urban India.
- Built an emotion-card reflection system in Google Apps Script and Google Sheets with the GPT API. Participants combine card codes to generate a reflection prompt, and the system parses their reflection into emotions. Tested four response strategies: empathic generation, curation of philosophy and poetry, generated fables, and a reframe with action steps. Demonstrated the final version at DIS 2025.
- Adapted the system into an SPA reflection tool for older adults in India. Replaced the emotion cards with SPA cards and grounded responses in a curated database of Bhagavad Gita verses and interpretations.
- Worked with a collaborator at IIT Gandhinagar to build the Android app. Deployed it for six weeks with 17 older adults across four living contexts.

Visiting Scholar, Human-AI Interaction (HAIX) Lab, IIT Gandhinagar, India

Aug 2025 – Aug 2026

Host: Yogesh Kumar Meena

- Developed SPA design materials (reflection probe, empathic reflection cards, SPA aspiration cards) and studied their use with 10 HCI practitioners and 12 older adults.
- Co-organized a half-day workshop at IndiaHCI 2025: [Designing for Self-Perception of Aging in Low-Resource Settings: Leveraging Reflection, AI, and Cultural Contexts in HCI](#).

- Research Assistant**, MemeX Project, ITI/LARSyS, Instituto Superior Técnico, University of Lisbon 2020 – 2021
- Worked as UX designer for the MemeX mobile app in this EU Horizon 2020-funded project.
 - Conducted research on inclusive digital technologies, migrant communities, and digital storytelling.
- Research Assistant**, Center for Inclusive Design and Environmental Access, University at Buffalo 2018
- Ran a user study with 10 participants that informed the next design phase of Scan-sign, an NFC-based indoor navigation system for people with visual impairments.
- Volunteer Researcher**, Inclusive Design Research Centre, OCAD University 2015 – 2016
- Developed use cases, requirements, and wireframes, and co-developed an interactive Framer.js prototype for the MyLearning Lab project.

SKILLS

Prototyping: Processing, Arduino with IMU sensors, Google Apps Script and Google Sheets with LLM API integration.
Design: information architecture, interaction flows, requirements and scope definition, user scenarios and personas, interface design. Tools: Adobe Illustrator, Adobe Creative Suite, Procreate.
Research: Research through Design, cultural probes, co-design, design cards, qualitative interviews, reflexive thematic analysis.
Art: drawing, digital illustration, linocut printmaking, movement improvisation.

EXHIBITIONS AND INTERACTIVE INSTALLATIONS

khanujaneeta.com

Interactive Installations and Performance

Ambidextrous Motion. Urmila Art Gallery, Kanoria Centre for Arts, Ahmedabad 2026
 Interactive drawing installation in the Kanoria Centre for Arts annual group exhibition. Arduino-based IMU sensors worn on each hand translate two-handed drawing into a live digital trace. About 25 visitors took part in a two-hour session, each producing a paper drawing and a matching digital trace.

Body as a Canvas. Lisbon 2025
 Live body painting on movement artist Safire Hikari.

Short movement video for an exhibition by Hilde Pank. Germany 2021
 Shown at the exhibition finissage; also shown in KOSMOS Chaos, Schloss Gleina, Germany, 2024.

Lost in Place. Bauhaus-Universität Weimar 2016
 Interactive installation using live camera capture, motion tracking, projection, and Processing.

Solo Exhibitions

Where We Need to Be. 1399 Gallery, Lisbon 2025

Discover. Atelier da Graça, Lisbon 2024

Connect. Beryti Café, Barcelona 2023

Moments. Wick Monet Gallery, Pittsburgh 2023

Confined. Modern Art Gallery, New Delhi 2021

Group Exhibition, Residency, Commission, and Press

Annual Group Exhibition , Kanoria Centre for Arts, Urmila Art Gallery, Ahmedabad (paintings and <i>Ambidextrous Motion</i>)	2026
Artist Residency , Kanoria Centre for Arts, Ahmedabad	2026
Commission : Album cover artwork for <i>Clouds</i> by The Ballroom Thieves (Netzwerk Music Group). Commission completed in 2021; album released in 2022.	2021
Press : <i>Divya Bhaskar</i> , Ahmedabad City edition (in Gujarati), coverage of <i>Ambidextrous Motion</i>	Feb 2026

AWARDS, FUNDING, AND FELLOWSHIPS

Best Paper Award (co-author) , ACM CHI 2026 (publication [2])	2026
Doctoral Funding , Fundação para a Ciência e a Tecnologia (FCT), Portugal	2021 – Present
Best Student Award , International Media Architecture Master Studies (highest grade)	2019
DAAD Funding , German Academic Exchange Service, dual master's program	
STIBET Scholarship , Bauhaus-Universität Weimar	
Fellowship , Neudeli Business Incubator, Bauhaus-Universität Weimar	

TEACHING

Teaching Assistant , Introduction to Accessibility and Assistive Technology (05-332/05-632) <i>Human-Computer Interaction Institute, Carnegie Mellon University. Instructor: Andrew Beigel</i>	Fall 2026
Teaching Assistant , User-Centered Design (CCU) <i>Instituto Superior Técnico, University of Lisbon. Instructors: Nuno Jardim Nunes and Valentina Nisi</i>	Fall 2021

DESIGN AND INDUSTRY EXPERIENCE

User Experience Design Intern , Bettzeit Gruppe, Frankfurt, Germany Built a customer experience map linking management, marketing, and development teams; designed UX for an e-commerce rebrand.	2018 – 2019
Venture and Design Coach , Blackstone LaunchPad, University at Buffalo Mentored early-stage startups on value propositions, design process, and MVP development.	2017 – 2018
Co-founder and Principal Designer , Algo Design Studio, Ahmedabad, India Led UX, UI, and interaction design for clients in India, Germany, and Russia; managed a four-person team.	2012 – 2016
Interaction Design Consultant , Emotome / NDBI, Ahmedabad, India Led design research, concept development, prototyping, and testing for the Gesture Library project.	2011 – 2012
Interaction Design Intern , Experiential Design Lab, Delhi, India Conducted user research and designed interaction flows for interactive installations and games.	2009 – 2010
Software Developer , Gateway Technolabs, Ahmedabad, India Developed web components in MXML and ActionScript 2.0/3.0.	2007 – 2008

ADDITIONAL TRAINING

Diploma , Dance Movement Therapy, Sancheti Healthcare Academy, Pune, India	2020 – 2021
500-Hour Yoga Teacher Training , Gyanish International Academy of Yogic Science, India	2021
Diploma in Movement Arts and Mixed Media (Honors) , Attakalari Centre for Movement Arts, India	2019 – 2020